

The concept of a vector

A quantity that needs a direction as well as a size — and the notation that carries both.

LESSON 39

1 × 40 MIN

GEOMETRY 8, TEMA 35

STAGE 9 · 13.1

LESSON CLOCK

5'

12'

12'

7'

4'

Which of these need a direction?

5 min

Vectors and scalars

12 min

Length and notation

12 min

Equal and opposite

7 min

Homework

4 min

LEARNING OBJECTIVES

- Distinguish a vector from a scalar.
- Use the notations $\overrightarrow{AB}$ with an arrow, and $\mathbf{a}$ in bold.
- Find the length (modulus) of a vector.
- Recognise equal, opposite and collinear vectors.

TEXTBOOK

National

Tema 35, pp. 78–80

Cambridge

Learner’s Book pp. 284–290

Workbook

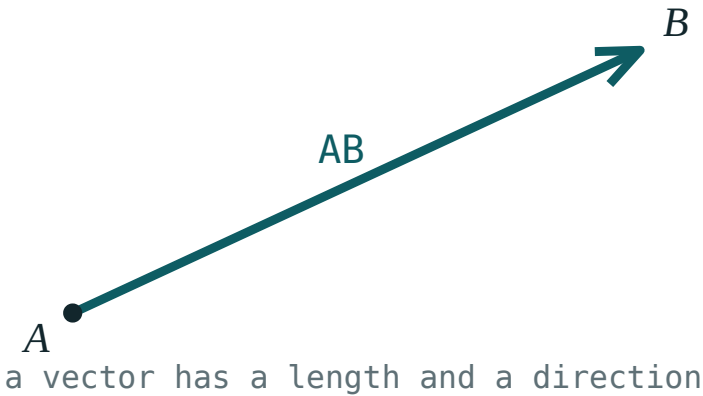
Workbook 13.1

01

Explanation

Size is not always enough

“The bus travelled 40 km” is a **scalar** — one number. “The bus travelled 40 km north” is a **vector**: a size *and* a direction. Force, velocity and displacement are vectors; mass, time, temperature and area are scalars.



A vector is a directed segment: it has a length and a direction, but no fixed position.

A vector is drawn as an arrow from its **initial point** to its **terminal point** and written AB with an arrow over it, or as a single bold letter $\mathbf{a}$.

LENGTH OF A VECTOR

If $\mathbf{a} = (a_1; a_2)$ then

$$|\mathbf{a}| = \sqrt{a_1^2 + a_2^2}$$

— the distance formula again. A length is a scalar, and it is never negative.

Equal, opposite, collinear

RELATIONSHIP	MEANS	IN COORDINATES
equal	same length, same direction	$a_1 = b_1$ and $a_2 = b_2$
opposite	same length, reversed direction	$(-a_1; -a_2)$
collinear	parallel — same or opposite direction	one is a multiple of the other
zero vector	no length, no direction	$(0; 0)$

A VECTOR HAS NO ADDRESS

Two arrows of the same length pointing the same way are the **same vector**, even in different corners of the page. That freedom is exactly what makes vectors useful — you may slide one anywhere to add it to another.

02 Worked examples

EXAMPLE 1 Find the vector AB and its length, for $A(1; 2)$ and $B(5; 5)$.

① $AB = (5 - 1; 5 - 2)$
End minus start — always in that order.

② $AB = (4; 3)$

③ $|AB| = \sqrt{16 + 9} = 5$

ANSWER

$(4; 3)$, length 5

EXAMPLE 2 Are $a = (2; -3)$ and $b = (-6; 9)$ collinear?

① $-6 = -3 \cdot 2$
Compare the first coordinates.

② $9 = -3 \cdot (-3)$
The same multiplier works for the second.

③ $b = -3a$
One is a multiple of the other.

④ They are collinear, pointing in opposite directions.
The multiplier is negative.

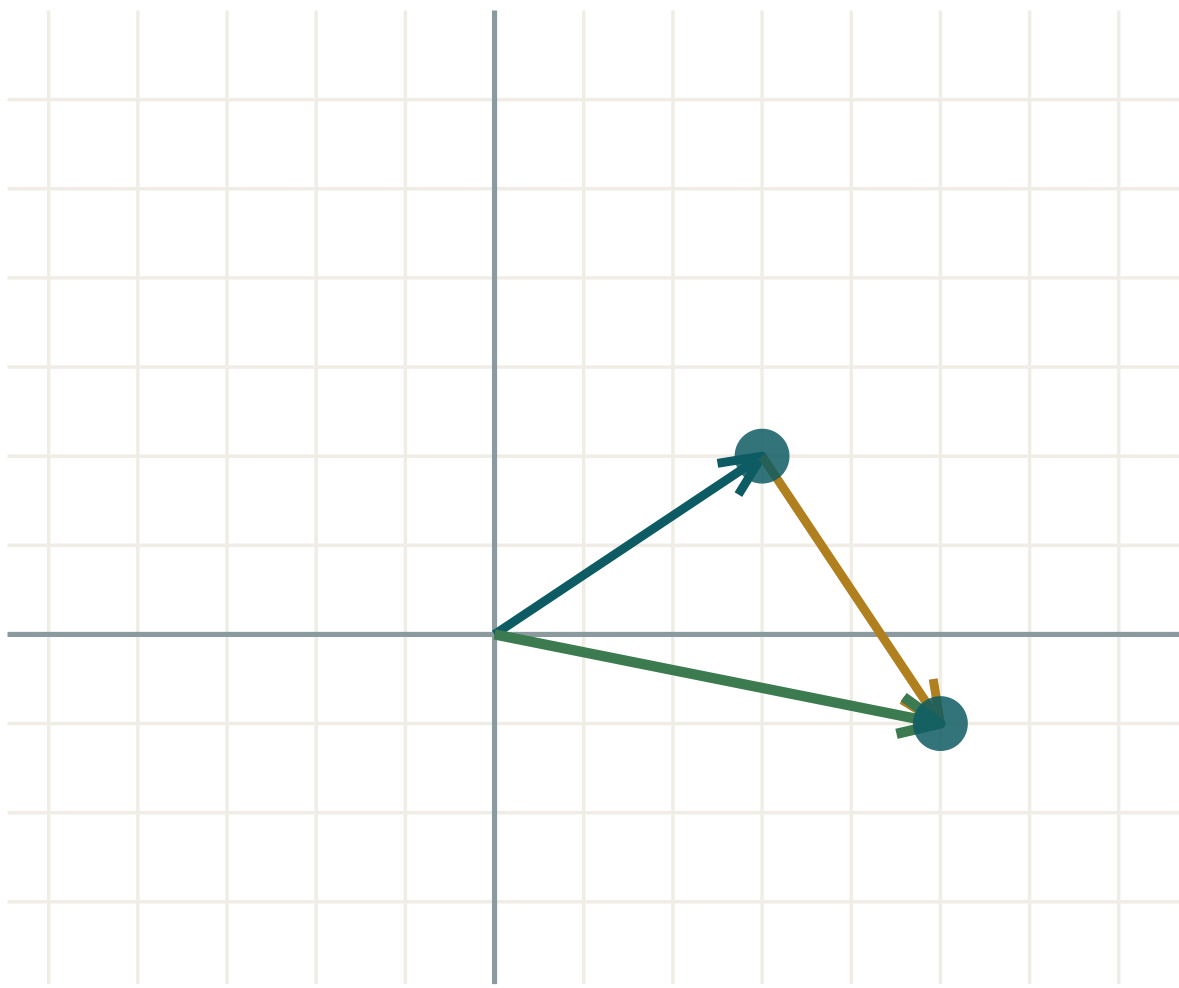
ANSWER

Yes — $b = -3a$

03 Interactive model

Drag the tip of the arrow and watch the coordinates and the length change together.

A vector and its length


 $a \ (3; 2)$
 $b \ (2; -3)$
 $a + b \ (5; -1)$
 $|a| \ 3.61$
 $|b| \ 3.61$
 $|a + b| \ 5.1$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Vector	Vektor	Вектор
Scalar	Skalyar	Скаляр
Directed segment	Yo'naltirilgan kesma	Направленный отрезок
Initial point	Boshi	Начало вектора
Terminal point	Oxiri	Конец вектора
Modulus (length)	Vektor uzunligi (moduli)	Модуль (длина) вектора
Equal vectors	Teng vektorlar	Равные векторы
Opposite vector	Qarama-qarshi vektor	Противоположный вектор
Zero vector	Nol vektor	Нулевой вектор
Collinear vectors	Kollinear vektorlar	Коллинеарные векторы

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Which of these is a vector?

mass

temperature

velocity

area

For $A(2; 1)$ and $B(6; 4)$, AB is:

$(4; 3)$

$(-4; -3)$

$(8; 5)$

$(3; 4)$

The length of $(-3; 4)$ is:

1

5

7

-5

The vector opposite to $(5; -2)$ is:

$(-5; 2)$

$(2; -5)$

$(5; 2)$

$(-5; -2)$

$(1; 2)$ and $(3; 6)$ are:

equal

opposite

collinear

perpendicular

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find AB for $A(0; 0)$, $B(3; 4)$.

ANSWER $(3; 4)$

2. Find the length of $(6; 8)$.

ANSWER 10

3. Find the length of $(0; -7)$.

ANSWER 7

4. Write the vector opposite to $(2; 5)$.

ANSWER $(-2; -5)$

5. Is force a vector or a scalar?

ANSWER A vector.

6. Find AB for $A(1; 1)$, $B(1; 6)$.

ANSWER $(0; 5)$

7. Find the length of $(-1; 0)$.

ANSWER 1

Medium · the standard question

7 PROBLEMS

1. Find AB and $|AB|$ for $A(-2; 3)$, $B(4; -5)$.

ANSWER $(6; -8)$, length 10

2. Are $(4; 6)$ and $(6; 9)$ collinear?

ANSWER Yes — $(6; 9) = 1.5(4; 6)$

3. Are $(2; 3)$ and $(3; 2)$ collinear?

ANSWER No — no single multiplier works.

4. Find the length of $(-5; 12)$.

ANSWER 13

5. $B(4; 7)$ and $AB = (1; 3)$. Find A .

ANSWER $A(3; 4)$

6. Find the exact length of $(2; 3)$.

ANSWER $\sqrt{13}$

7. Write a vector equal to AB where $A(0; 0)$, $B(2; -1)$, starting at $(5; 5)$.

ANSWER From $(5; 5)$ to $(7; 4)$

Hard · stretch and reason

7 PROBLEMS

1. $|a| = 5$ and $a = (3; k)$. Find all k .

ANSWER $k = \pm 4$

2. Find a vector of length 10 collinear with $(3; 4)$.

ANSWER $(6; 8)$ or $(-6; -8)$

3. Show that $A(1;1)$, $B(4;5)$, $C(7;9)$ are collinear using vectors.

ANSWER $AB = (3; 4)$, $BC = (3; 4)$ — equal vectors, so the three points lie on one line.

4. $|a| = 13$ and $a = (k; 12)$. Find all k .

ANSWER $k = \pm 5$

5. Find a unit vector (length 1) in the direction of $(6; 8)$.

ANSWER $(0.6; 0.8)$

6. $A(2; 1)$, $B(6; 4)$, $C(9; 8)$. Is ABC a straight line?

ANSWER No — $AB = (4; 3)$ and $BC = (3; 4)$ are not multiples of each other.

7. Explain why a vector has no fixed position but a directed segment does.

ANSWER A vector records only length and direction; any two arrows agreeing on both represent the same vector, while a directed segment also carries its start point.

07 Homework

Homework — 5 problems

Geometry 8, Tema 35, pp. 78–80. Draw every vector as an arrow on squared paper.

- Find AB and $|AB|$ for $A(1; 2)$, $B(7; 10)$.
- Find the length of $(-8; 15)$.
- Write the vector opposite to $(-4; 9)$.
- Are $(2; -6)$ and $(-1; 3)$ collinear? Justify.

5. Name three vector quantities and three scalar quantities from physics.